

1 A regular pentagon with a centrally symmetric partner

The preceding result allows the two pentagons to deform independently. We now keep one regular pentagon fixed and allow its partner to be any centrally symmetric planar convex body, including a smooth body. The capacity is then determined by a single convex-body inclusion. Its sharp systolic ratio is strictly below one.

Use the preceding normal convention, writing

$$n_i = (\cos(2\pi i/5), \sin(2\pi i/5)), \quad Q = \{p : n_i \cdot p \leq h, \ 0 \leq i < 5\}, \quad h = \cos(\pi/5).$$

Thus Q has circumradius one. For a convex body A containing the origin in its interior, write $A^\circ = \{x : x \cdot y \leq 1 \text{ for every } y \in A\}$. Define

$$B = \frac{1}{2}(Q - Q)^\circ = \text{conv} \left\{ \frac{\pm n_i}{2(1+h)} : 0 \leq i < 5 \right\}, \quad W(K) = \max\{r \geq 0 : r(Q - Q)^\circ \subseteq K - K\}.$$

The body B is a regular decagon with its orientation fixed by Q . The product below is Lagrangian, and c denotes its Ekeland–Hofer–Zehnder capacity.

Theorem 1. *For every centrally symmetric planar convex body K with nonempty interior,*

$$c(K \times_L Q) = W(K), \quad \frac{c(K \times_L Q)^2}{2 \text{area}(K) \text{area}(Q)} \leq \kappa_{\text{sym}} := \frac{10 + 6\sqrt{5}}{25} < 1.$$

Equality in the ratio holds exactly when K is a translate of a positive dilate of B .

The proof uses a translation-cover characterization of capacity. For an oriented polygonal curve with edge vectors v_j , its Q -length is $\sum_j s_Q(v_j)$, where $s_Q(v) = \max_{p \in Q} v \cdot p$. This length can depend on orientation, since Q is not symmetric. Balitskiy–Mitrofanov–Polyanskii show that $c(K \times_L Q) \geq 1$ exactly when every closed curve of unit Q -length fits into a translate of K [BMP, Theorem 2.5]. For polygonal Q , it suffices to check the *normal triangles*: unit-length oriented triangles whose edges follow outward normal directions of Q , including degenerate triangles when opposite normals exist [BMP, Definition 3.2 and Theorem 3.3]. Our pentagon has no opposite normals, so all its normal triangles are nondegenerate. There are ten translation classes. Their underlying triangles have angles $\pi/5, \pi/5, 3\pi/5$ and are rotations of one another through multiples of $\pi/5$ [BMP, Section 6.1]. For odd multiples, the traversal orientation reverses so that the edges again follow outward normals.

Proof. We first compute $c(B \times_L Q)$. Put $\theta = \pi/5$ and $R = 1/[2(1+h)]$, the circumradius of B . If a normal triangle has longest side L , its other sides both have length $L/(2 \cos \theta)$. Its oriented edges follow outward normals, so their Q -lengths equal h times their Euclidean lengths. Consequently

$$1 = hL(1 + 1/\cos \theta) = L(1 + h), \quad L = 2R.$$

Translate the longest side's midpoint to the origin. Its endpoints are antipodal vertices of B , and the third vertex has distance $R \tan \theta$ from the origin. This is smaller than the incircle radius $R \cos(\theta/2)$, because

$$\tan^2(\pi/5) = 5 - 2\sqrt{5} < \frac{5 + \sqrt{5}}{8} = \cos^2(\pi/10).$$

Each of the ten triangles therefore lies in B , giving $c(B \times_L Q) \geq 1$.

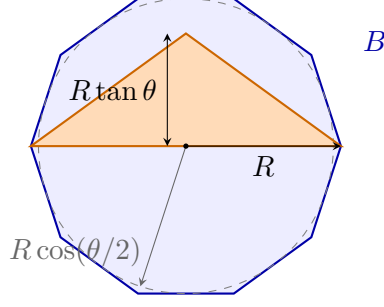


Figure 1: A translated normal triangle in B . Its longest side joins antipodal vertices; its third vertex lies inside the dashed incircle. The same placement works after each rotation by $\pi/5$.

For any partner K , the curve-cover criterion gives the opposite bound $c(K \times_L Q) \leq W(K)$. Indeed, a segment traversed out and back with displacement v has length $s_Q(v) + s_Q(-v) = s_{Q-Q}(v)$. It fits into a translate of K exactly when $v \in K - K$. If capacity is at least $r > 0$, rescale the first factor by $1/r$ and apply the criterion to all unit-length segments. Convexity of $K - K$ then gives $r(Q - Q)^\circ \subseteq K - K$. Since $B - B = (Q - Q)^\circ$, this proves $c(B \times_L Q) = W(B) = 1$.

Now translate a symmetric K so that $K = -K$. Then $K - K = 2K$, and the definition of $W(K)$ gives $W(K)B \subseteq K$. Monotonicity and homogeneity in one factor yield $c(K \times_L Q) \geq W(K)c(B \times_L Q) = W(K)$, proving the capacity formula.

The same containment gives $\text{area}(K) \geq W(K)^2 \text{area}(B)$. Proper inclusion between full-dimensional convex bodies strictly increases area, so equality holds exactly for translates of $W(K)B$. Finally,

$$\text{area}(B) = 5R^2 \sin(\pi/5), \quad \text{area}(Q) = \frac{5}{2} \sin(2\pi/5), \quad \frac{1}{2 \text{area}(B) \text{area}(Q)} = \frac{10 + 6\sqrt{5}}{25}.$$

This gives the sharp ratio and its equality class. $\square$

The formula also identifies the bound W for an asymmetric partner. Its central symmetrization $S = (K - K)/2$ satisfies $S - S = K - K$, hence

$$W(K) = c(S \times_L Q), \quad c(K \times_L Q) \leq c(S \times_L Q).$$

Thus the quotient $c(K \times_L Q)/W(K)$ records the fraction of the symmetrized capacity retained by K . In particular, a counterexample with this fixed pentagon factor must use an asymmetric partner. The argument explains this restriction by calibrating one reference cover and transferring its capacity through containment.

References

- [BMP] A. Balitskiy, I. Mitrofanov and A. Polyanskii, *Triangle covering problems and the Viterbo inequality in the plane*, arXiv:2603.12495v2, 19 August 2026. Source with theorem numbering.